MATH 345 Skeleton Notes

Unit 7: Abstract vector spaces and polynomial approximation

Section 7.1: Abstract vector spaces

Earlier units worked mostly with vectors in $\mathbb{R}^n$. The same algebraic rules also make sense for matrices and polynomials. This section gives the abstract definition so that later courses can use the same language in many settings: vectors may be columns, matrices, polynomials, or other objects.

Definition A real $\boxed{\text{vector space}}$ consists of a nonempty set V of objects (which we will call $\boxed{\text{vectors}}$) that can be added together, and multiplied by a real number (called a $\boxed{\text{scalar}}$ in this context) and for which the ten axioms below hold. If $\mathbf{v}$ and $\mathbf{w}$ are two vectors in V, their sum is denoted $\mathbf{v} + \mathbf{w}$, and the scalar product of $\mathbf{v}$ by a real number a is denoted by $a\mathbf{v}$. The axioms of a vector space are described below:

(A1) Closure Under Addition: If $\mathbf{u}$ and $\mathbf{v}$ are any elements in V, then $\mathbf{u} + \mathbf{v}$ is in V.

(A2) Commutativity of Addition: For any $\mathbf{u}, \mathbf{v} \in V$, $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$.

(A3) Associativity of Addition: For any $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$, $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$.

(A4) Existence of a Zero Vector: There exists a $\boxed{\text{zero vector}}$ $\mathbf{0}$ in V, such that $\mathbf{u} + \mathbf{0} = \mathbf{0} + \mathbf{u} = \mathbf{u}$ for any $\mathbf{u} \in V$.

(A5) Existence of Additive Inverses For each $\mathbf{v} \in V$ there exists $-\mathbf{v}$ in V such that

$$\mathbf{v} + (-\mathbf{v}) = -\mathbf{v} + \mathbf{v} = \mathbf{0}.$$

The vector $-\mathbf{v}$ is the *negative* or

additive inverse of $\mathbf{v}$.

(S1) Closure Under Scalar Multiplication If $\mathbf{v}$ is any element of V and a is any real number, then $a\mathbf{v}$ is in V.

(S2) Distributive Law 1 For any $\mathbf{u}, \mathbf{v} \in V$ and any $a \in \mathbb{R}$, $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$.

(S3) Distributive Law 2 For any $\mathbf{v} \in V$ and any $a, b \in \mathbb{R}$, $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$.

(S4) Associativity of Scalar Multiplication For any $\mathbf{v} \in V$ and $a, b \in \mathbb{R}$, $a(b\mathbf{v}) = (ab)\mathbf{v}$.

(S5) $1\mathbf{v} = \mathbf{v}$ for any $\mathbf{v} \in V$.

The operation $\mathbf{v} + \mathbf{w}$ is called

vector addition and the operation $a\mathbf{v}$ is called

scalar multiplication.

Note In order to be clear about the operations used, we will often denote newly defined addition and scalar multiplication operations using the symbols $\oplus$ and $\odot$, respectively.

Note We often use the phrase "vector space", dropping the "real" and assuming our scalars are real numbers in this course. However, it is possible to define complex vector spaces, rational vector spaces, and more, using different systems of scalars than the real numbers.

Note From now on, a "vector" will mean an element of a vector space, not necessarily a column vector or line segment.

The main examples in this unit are

$$\mathbb{R}^n, \qquad M_{m,n}, \qquad \mathscr{P}_{\le d}.$$

Here $M_{m,n}$ is the set of $m \times n$ matrices, and

$$\mathscr{P}_{\le d} = \{a_0 + a_1 x + \cdots + a_d x^d \, : \, a_0, \ldots, a_d \in \mathbb{R}\}.$$

In $\mathbb{R}^n$, addition and scalar multiplication are entrywise. In $M_{m,n}$, they are entrywise. In $\mathscr{P}_{\le d}$, they are polynomial addition and scalar multiplication. The set $\mathscr{P}$ of all polynomials is also a vector space, but it is infinite-dimensional and will not be our main example.

Activity

Consider the vector space $\mathbf{M}_{22}$ of all 2×2 matrices.

Task Prove that $\mathbf{M}_{22}$ is closed under addition and scalar multiplication. What is the zero vector in this vector space?

Task What is the zero vector in $\mathbf{M}_{22}$?

Activity

Let's walk through the proof that $\mathscr{P}_{\le 2}$ is a vector space.

Activity

Let V be the set of all ordered pairs of real numbers (x, y) with the operations

$$(x_1, y_1) \oplus (x_2, y_2) = (x_1, 0) \quad \text{and} \quad c \odot (x, y) = (cx, cy).$$

Show that V is *not* a vector space.

Activity

Let V be the set of all ordered pairs of real numbers (x, y) with the operations

$$(x_1, y_1) \oplus (x_2, y_2) = (x_1 + x_2, y_1 + y_2) \quad \text{and} \quad c \odot (x, y) = (cx, y).$$

Show that V is not a vector space.

Activity

Let V be the set of all non-negative real numbers, and define vector addition as

$$\mathbf{v} \oplus \mathbf{u} = |\mathbf{v} - \mathbf{u}|,$$

and scalar multiplication as

$$a \odot \mathbf{v} = |a|\mathbf{v}.$$

Does this structure form a vector space?

Below we list several facts about the interaction of addition and multiplication that hold in any vector space.

Theorem Let V be a vector space. For any $\mathbf{v} \in V$ and $a \in \mathbb{R}$,

- $0\mathbf{u} = \mathbf{0}$.
- $a\mathbf{0} = \mathbf{0}$.
- If $a\mathbf{v} = \mathbf{0}$, then $a = 0$ or $\mathbf{v} = \mathbf{0}$.
- $(-1)\mathbf{v} = -\mathbf{v}$.
- $(-a)\mathbf{v} = -(a\mathbf{v}) = a(-\mathbf{v})$.

The point is not that every vector looks like an arrow. The point is that the same algebraic rules let us reuse span, independence, bases, coordinates, dimension, and linear maps in settings where the objects are matrices or polynomials.